

COMPSCI 773: Camera Calibration

Patrice Delmas¹

School of Computer Science, The University of Auckland

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¹with material from Mitchell Rogers, Shahrokh Heidari, Tony Wang, Ruigeng Wang, and Gibran Zazueta

Overview of the stereo pipeline

Step 1: Camera selection



Step 2: Capture images from each camera with inherent optical distortion

Camera 1 (distorted)



Camera 2 (distorted)



Step 3: Distortion removal (undistorted images)

Camera 1 (undistorted)



Camera 2 (undistorted)



Step 4: Mount cameras to create Stereo Rig



Step 5: Intrinsic calibration of camera one and two

Cam 1:
Focal length = 7315.25
Optical center: (997, 1806)
Distortion = 0.0003

Cam 2:
Focal length = 7317.25
Optical center: (997, 1806)
Distortion = 0.0003

Step 6: Extrinsic calibration of the Stereo System

Translation vector: [-99.84, 0.822, 0.436]^T

Rotation matrix: $\begin{bmatrix} 0.999 & 0.004 & 0.004 \\ -0.004 & 0.999 & -0.006 \\ -0.006 & 0.006 & 0.999 \end{bmatrix}$

Step 7: Stereo rectification



Step 8: Compute disparity image



Step 9: Calculate depth



Step 10: Create 3D object



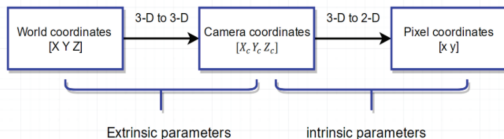
Figure: Overview of the steps in the stereo pipeline.

The Binocular Stereo Pipeline

- 1 **Image acquisition and Distortion removal:** Acquire the input images and warp the images to remove all optical distortion.
- 2 **(Stereo) camera calibration:** Define the pinhole model and compute all intrinsic and extrinsic calibration parameters mapping 2D image points to 3D scene points.
- 3 **Rectification:** Warp the images so that the rules of canonical epipolar geometry can be applied to the images.
- 4 **Stereo matching:** Find corresponding points between the images, and store these correspondences as disparities in a disparity map.
- 5 **Point cloud:** Convert disparity values into depth values. Find 3D locations of 2D points. Write these values into a PLY file.

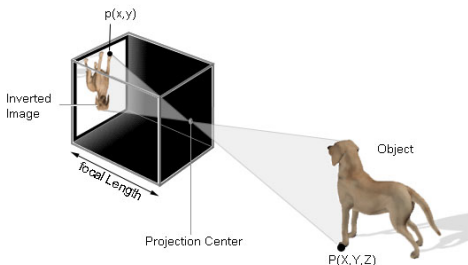
Camera Calibration

- Calibrating a camera means determining the imaging process parameters that map a 3D point in a reference frame onto its 2D image position in pixel unit coordinates.
- Parameters:
 - External, or **extrinsic**, parameters relate to a transformation of the World Reference Frame (WRF) into the Camera Reference Frame (CRF). Extrinsic parameters describe the camera's position and orientation in 3D space.
 - Internal, or **intrinsic**, parameters relate to the optics and the physics of the camera. Intrinsic parameters refer to the camera's internal characteristics, such as focal length, principal point, and distortion coefficients



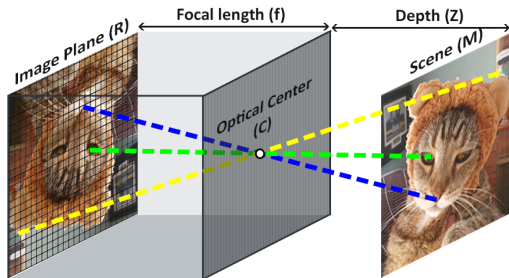
Pinhole camera model

- The pinhole camera model is a simplified mathematical model that describes the projection of a 3D scene onto a 2D image plane through a small aperture (or pinhole).
- Through the pinhole, some of the light rays emitted or reflected by objects form an inverted image of these objects on the back screen. The operation, which creates this inverted image, is called a **perspective projection**.
- The model assumes that light travels in straight lines and that objects are imaged as points onto the image plane.



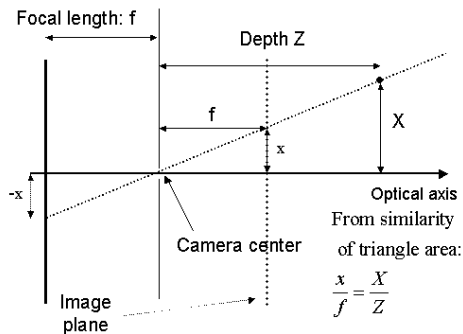
Geometric pinhole model

- The pinhole camera model defines a set of intrinsic parameters that characterize the camera's internal geometry, such as the focal length, the principal point (the intersection of the image plane and the optical axis), and the aspect ratio of the image sensor.
- An image is formed in the image plane R through a perspective projection. The optical centre C at a distance f (the focal length of the optical system). A 3D point M at a depth Z from the optical centre C is projected onto R by intersecting the ray CM with R . Then, an inverted image of the object is formed.



Geometric Pinhole Model

The pinhole camera model



Traditionally, the image plane is drawn in front of the focal point.

Transforming 3D points into 2D image points

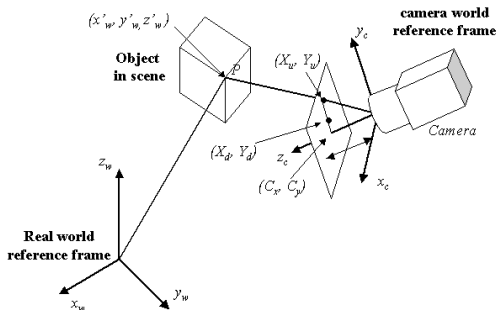
Given 3D points M_{3D} in the world and 2D image points M_I , we are aiming to find a Transform T such that: $M_I = TM_{3D}$

- In WRF, a point is measured in μm , mm, inches, light years, and parsecs.
- In the image reference frame, points are measured in pixels, which is the size of a photosensitive camera receptor.
- We need transformations (e.g., rotation, translation, scaling, affine transforms) to express/transform coordinates of 2D and 3D points in different reference frames (with respect to the world and with respect to the camera).

Transforming 3D points into 2D image points: monocular camera calibration

What is required to perform monocular camera calibration?

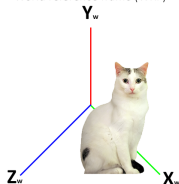
- Coordinate systems (World, Camera, Image) and transformations between them
- 3D to 2D points projection



Coordinate systems: From the world to the camera

World Coordinates

World reference frame (WRF)

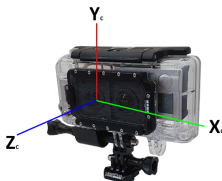


Units: cm, mm, inches etc

Each point can be represented by a 3D vector with an (x, y, z) position relative to some known point (World origin)

Camera Coordinates

Camera reference frame (CRF)

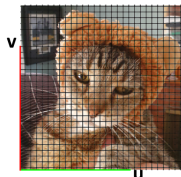


Units: cm, mm, inches etc

Each point can be represented by a 3D vector with an (x, y, z) position relative to the camera (Camera origin)

Image Coordinates

Image plane coordinates



Units: pixels

Each point is given by its 2D pixel coordinate

Figure: Three coordinate systems we use in 3D computer vision.

Mathematical Requirements for Monocular Camera Calibration

- A matrix framework to easily stack affine transforms (homogeneous coordinates)
- 3D to 3D transforms (rotation-translation-scaling)
- 3D to 2D transform (perspective projection)
- 2D to 2D transform (rotation-translation-scaling)
- Distortion removal (non-linear process)
- Refinement process (Optimization)
- Error estimation (line to plane intersection, back projection, projection)
- Linear algebra operations

We need to review 2D and 3D geometry basics!!

Homogeneous coordinates of a 1D point -1

- In inhomogeneous 1D (Cartesian) coordinates, a point is represented by a single value (for example, $x = 1$).
- In homogeneous coordinates, a 1D point is represented by any scaled version of a 2D vector:

$$\begin{bmatrix} x \\ 1 \end{bmatrix}$$

For example:

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \sim \begin{bmatrix} 4 \\ 2 \end{bmatrix} \sim \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

Homogeneous Coordinates of a 1D Point - 2

All scaled vectors represent the same Cartesian point:

$$x = \frac{x_h}{w}$$

where:

- x_h is the homogeneous coordinate
- w is the scale factor

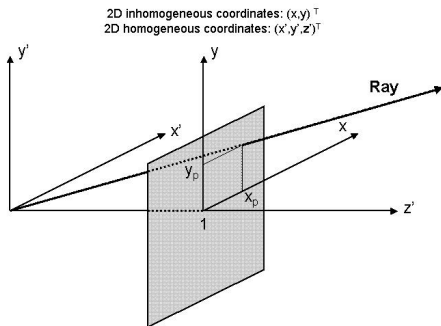
Homogeneous coordinates are useful because:

- translations become matrix multiplications
- perspective projection can be represented using matrices

2D Homogeneous coordinates

In inhomogeneous 2D (Cartesian) coordinates, a point is represented by a 2D vector $(x, y)^T$. In homogeneous coordinates, it is viewed as a 3D

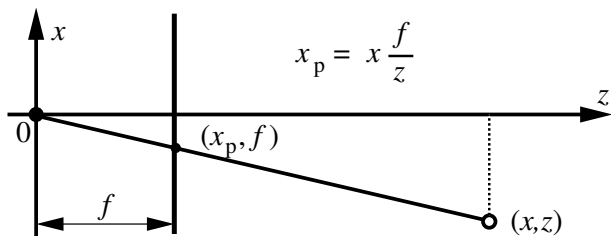
vector $\begin{pmatrix} x_h \\ y_h \\ z_h \end{pmatrix}$ or multiple of the vector $\begin{pmatrix} \frac{x_h}{z_h} \\ \frac{y_h}{z_h} \\ 1 \end{pmatrix}$.



Notation Summary

Symbol	Coordinate Space	Meaning
(X, Y, Z)	World Reference Frame	3D world coordinates
(x, y, z)	Camera Reference Frame	3D camera coordinates
(x_p, y_p)	Image Plane	Coordinates in mm
(u, v)	Image Coordinates	Pixel coordinates
f	Camera Model	Focal length (mm)
f_x, f_y	Intrinsic Matrix	Focal length (pixels)
(c_x, c_y)	Intrinsic Matrix	Principal point
R, t	Extrinsic Parameters	Rotation and translation
K	Intrinsic Matrix	Camera internal parameters
P	Projection Matrix	Full camera matrix

Projection onto a plane



Perspective Projection onto a Plane

Consider a 2D point P in the camera reference frame projected onto an image plane located at focal distance f :

$$P = \begin{bmatrix} x \\ z \end{bmatrix}$$

Using similar triangles: $x_p = f \frac{x}{z}$

Assuming $s = z$, the projection can be written in homogeneous coordinates as:

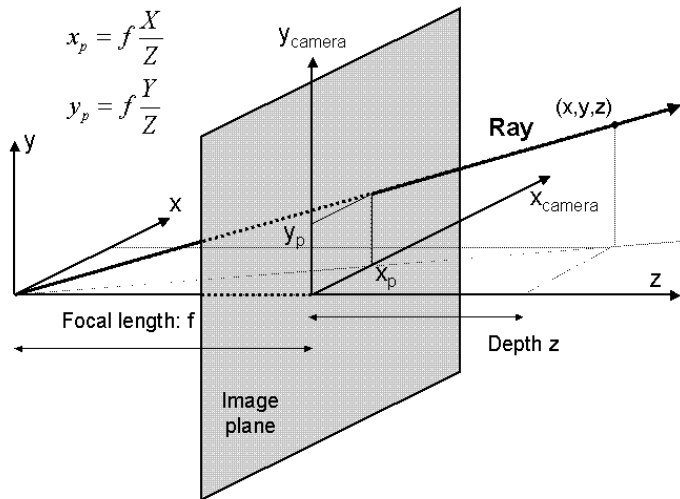
$$\begin{bmatrix} sx_p \\ s \end{bmatrix} = \begin{bmatrix} f & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ z \end{bmatrix}$$

Dividing by $s = z$ yields:

$$x_p = f \frac{x}{z}$$

This corresponds to the perspective projection of a 2D point onto the image plane.

Projection into the image plane



Perspective Projection onto the Image Plane

Using similar triangles, A 3D point in the camera reference frame is projected onto the image plane located at focal distance f :

$$P = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$x_p = f \frac{x}{z}, \quad y_p = f \frac{y}{z}$$

where:

- (x, y, z) are the coordinates of the 3D point in the camera reference frame
- (x_p, y_p) are the coordinates on the image plane (in mm)
- f is the focal length

Perspective Projection onto the Image Plane

The projection from the 3D camera reference frame to the image-centered coordinates (in *mm*) can be written in homogeneous coordinates as:

$$\begin{bmatrix} sx_p \\ sy_p \\ s \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} fx \\ fy \\ z \end{bmatrix}$$

with:

$$s = z$$

Dividing by $s = z$ yields:

$$x_p = f \frac{x}{z}, \quad y_p = f \frac{y}{z}$$

The Intrinsic Camera Matrix

The intrinsic matrix maps image plane coordinates (in mm) into pixel coordinates:

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

where: $f_x = \frac{f}{d_x}$ is the focal length in pixels along x , $f_y = \frac{f}{d_y}$ is the focal length in pixels along y , (c_x, c_y) is the principal point, and d_x, d_y is pixel size in mm in the x - and y - directions.

The intrinsic matrix combines the above parameters, and the projection into image coordinates becomes:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Image processing specifications

In image processing, the origin of the reference frame can be set in two different ways:

- Case 1: At the top left of the image, x -axis pointing rightward, y -axis downwards.
- Case 2: At the bottom left of the image, x -axis pointing rightward, y -axis upwards.
- Pixels are the unit of measurement: A pixel unit corresponds to the size of the photosensitive receptors, which transform the light rays into electrical or logical signals. They are not necessarily square.
- mm can be used to provide measurements in the image. It requires knowing the pixel size, usually specified in microns. It can be found online for standalone cameras or cameras on a certain type of phone.

Image processing specifications: Case 1

Transforming 2D points (in mm units) from a reference frame with the image center as origin into 2D points (in pixel units) in a reference frame of type 1, with image width W and height H is given by:

$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{d_x} & 0 & c_x \\ 0 & -\frac{1}{d_y} & c_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_p \\ y_p \\ 1 \end{pmatrix}$$

where $c_x = W/2$ and $c_y = H/2$

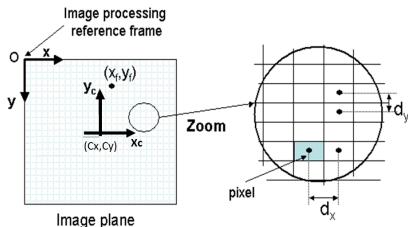
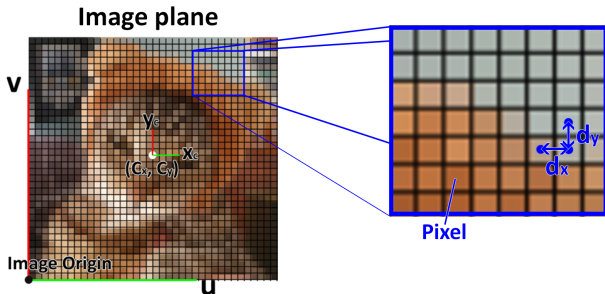


Image processing specifications: Case 2

Transforming 2D points (in mm units) from a reference frame with the image center as origin into 2D points (in pixel units) in a reference frame of type 2 is given by:

$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{d_x} & 0 & c_x \\ 0 & \frac{1}{d_y} & c_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_p \\ y_p \\ 1 \end{pmatrix}$$



Camera calibration - step 1: 3D to 3D Transform Matrix

A 3D point from the WRF, $(X, Y, Z, 1)^T$ in homogeneous coordinates, is transformed into its counterpart $(x, y, z, 1)^T$ in the CRF:

$$\begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} R_{3 \times 3} & T_{3 \times 1} \\ 0_{1 \times 3} & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

The 4×4 transform matrix combines rotation and translation in homogeneous coordinates. The matrices R and T describe the camera's position and orientation with respect to the new world reference frame. Their coefficients are called the **extrinsic parameters** of the camera.

Camera calibration - step 2: Projection into the image plane

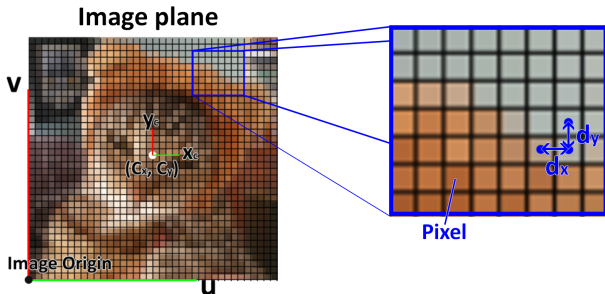
3D points in the CRF are projected using the pinhole camera model into the image plane situated at focal distance f of the CRF. Homogeneous coordinates with respect to the centre of the image derived from the perspective geometry transform of a 3D point onto a plane is given by:

$$\begin{pmatrix} sx_p \\ sy_p \\ s \end{pmatrix} = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}, s \neq 0$$

Camera calibration step 3 - From centered Image coordinates in mm to pixels

Transforming 2D points (in mm units) from a reference frame with the image center as origin into 2D points (in pixel units) in a reference frame of type 2 is given by:

$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{d_x} & 0 & c_x \\ 0 & \frac{1}{d_y} & c_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_p \\ y_p \\ 1 \end{pmatrix}$$



Calibration Summary

We can use homogeneous coordinates to map the pinhole projection camera model of 3D points (real values) to the image plane (positive integer values)

- Transform the 3D points to the CRF (world units (3D, *mm*) to camera coordinates (3D, *mm*)).
- Project the points onto the image plane (camera coordinates (3D, *mm*) to metric image plane (2D, *mm*)).
- Map the points on the camera image plane to the image plane coordinates (metric image plane in *mm* to pixels).

$$\begin{pmatrix} su \\ sv \\ s \end{pmatrix} = \begin{pmatrix} \frac{1}{d_x} & 0 & c_x \\ 0 & \frac{1}{d_y} & c_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

Camera Calibration Procedure

- 1 Determine the 3×4 theoretical calibration matrix
 - Rigid body transform from the WRF to the CRF: (R, t) , where R is the rotation matrix, and t is the translation vector.
 - Projection onto the image plane: (f) , where f is the focal length
 - Integrating optics and physics specifications (d_x, d_y)
- 2 Obtain measurements for a set of real points and their corresponding pixel locations in the image plane.
- 3 Compute the calibration matrix coefficients (point correspondence).
- 4 Compute the intrinsic and extrinsic calibration parameters.
- 5 Estimate your calibration errors:
 - Error in the image
 - Error in the cube

Calibration Summary-2

The relationship between a 3D (world frame) point M and its projected 2D point p in the image and measured in pixel is given as $\mathbf{p} \sim \mathbf{M}$ with P , the 3×4 Projection matrix written as:

$$P = K[R|t]$$

where K is the camera calibration matrix for the intrinsic (internal) parameters, R representing the rotation matrix describing the orientation of the camera with respect to the WRF and t the translation vector representing the translation of the camera origin with respect to WRF.

$$\begin{pmatrix} su \\ sv \\ s \end{pmatrix} = \underbrace{\overbrace{\begin{pmatrix} \frac{1}{d_x} & 0 & c_x \\ 0 & \frac{1}{d_y} & c_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}}^K \overbrace{\begin{pmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix}}^{[R|t]}}_P \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

Camera calibration - step 4: Solving the system

All characteristics of the camera can be expressed by the following 3×4 matrix, which transforms a 3D point in the World reference Frame $(X, Y, Z)^T$ into its pixel counterpart $(u, v)^T$ in the image:

$$\begin{pmatrix} su \\ sv \\ s \end{pmatrix} = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}, \text{ where } s \neq 0$$

Unknowns are the elements of the matrix, which can be related to the intrinsic and extrinsic parameters of the camera. Let us rewrite the system as:

$$\begin{cases} su = aX + bY + cZ + d \\ sv = eX + fY + gZ + h \\ s = iX + jY + kZ + 1 \end{cases}$$

Solving the system

Solving for u and v : We first multiply both sides of the third equation by u , then

$$\begin{aligned} su &= iXu + jYu + kZu + u \rightarrow u = su - iXu - jYu - kZu \\ &\rightarrow u = aX + bY + cZ + d - iXu - jYu - kZu \end{aligned}$$

Same can be done for v . Therefore, we have

$$\begin{cases} u = aX + bY + cZ + d - iXu - jYu - kZu \\ v = eX + fY + gZ + h - iXv - jYv - kZv \end{cases}$$

This yields the following linear system for a given set of points M_i ,

$i \in 1, \dots, n$, whose 3D coordinates $(X_i, Y_i, Z_i)^T$ and corresponding 2D pixels (u_i, v_i) are known (or measured):

Solving the system

$$\underbrace{\begin{pmatrix}
 x_1 & y_1 & z_1 & 1 & 0 & 0 & 0 & 0 & -x_1 u_1 & -y_1 u_1 & -z_1 u_1 \\
 0 & 0 & 0 & 0 & x_1 & y_1 & z_1 & 1 & -x_1 v_1 & -y_1 v_1 & -z_1 v_1 \\
 x_2 & y_2 & z_2 & 1 & 0 & 0 & 0 & 0 & -x_2 u_2 & -y_2 u_2 & -z_2 u_2 \\
 0 & 0 & 0 & 0 & x_2 & y_2 & z_2 & 1 & -x_2 v_2 & -y_2 v_2 & -z_2 v_2 \\
 x_3 & y_3 & z_3 & 1 & 0 & 0 & 0 & 0 & -x_3 u_3 & -y_3 u_3 & -z_3 u_3 \\
 0 & 0 & 0 & 0 & x_3 & y_3 & z_3 & 1 & -x_3 v_3 & -y_3 v_3 & -z_3 v_3 \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots
 \end{pmatrix}}_{\text{3D Measurement matrix } A}
 \underbrace{\begin{pmatrix} a \\ b \\ c \\ d \\ e \\ f \\ g \\ h \\ i \\ j \\ k \end{pmatrix}}_h = \underbrace{\begin{pmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{pmatrix}}_{\text{2Dpointsb}}$$

Since many point correspondences are usually available, the system is overdetermined and is solved in the least-squares sense using the Moore–Penrose pseudo-inverse:

$$Ah = b \text{ and } h = (A^T A)^{-1} A^T b$$

Camera Calibration process steps

- Design a Calibration object with either planar or non-coplanar points.
- Geometry and measurements of the 3D points should be precisely determined.
- Extract (manually or automatically) feature points (patches, corners or predefined patterns) from the image.
- Do the calibration (Tsai method or else).
- Assess the quality of your calibration (discrepancy between the real 3D coordinates of test points and their computed coordinates after calibration).
- Refine your calibration process.

Tsai Calibration outputs

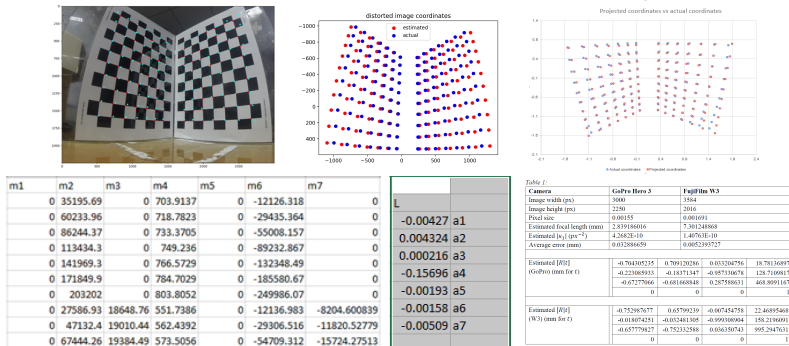


Figure: Calibration steps and results visualization examples